# Asad Ikram

CSH Graduate Program 2027, answers to the three questions · asad.ikram53@gmail.com


## 1. A research project, my contribution, and what did not go as expected

My MSc dissertation at the University of Portsmouth (2025, supervised by Dr Ella Haig) asked how public reaction to UK economic policy differed across social platforms. The project was mine from question to write-up. I obtained ethics approval, collected 255,446 public comments from X/Twitter, Reddit, YouTube and Quora through official APIs and licensed feeds, and built the analysis in Python: VADER and BiLSTM baselines, a fine-tuned BERT classifier, temperature-scaling calibration, McNemar and bootstrap tests, chi-square tests across topics, Prophet forecasts, NRC emotion scores, and LIME and integrated gradients to see what the model relied on. I also hand-labelled 200 posts myself as a check.

The part that did not go as expected was the validation. BERT reached a macro-F1 of 0.878, but its labels had been seeded by the VADER lexicon, the score was largely in-sample, and my 200-post check was a review of VADER's labels, so the Cohen's κ of about 0.78 measured agreement with VADER rather than between independent annotators. Every number pointed back at the same lexicon, so the validation was circular: it showed that BERT had learned to imitate VADER, not that either measured sentiment. Underneath sat a second assumption I had not tested: that a "negative" label means the same thing on Reddit as on YouTube, where people write for different audiences, at different lengths and under different norms.

I learned two things. A high agreement score describes the reference as much as the model, so the first design question is where the criterion comes from and whether it is independent of the thing being tested. And a comparison between groups needs evidence that the instrument behaves alike in each group, a routine test in survey research that computational text analysis rarely runs. With Dr Haig I turned this into a methods paper, *Is the Platform Part of the Measurement?*, which treats each platform as a survey mode and sets out a protocol built on generalizability theory, measurement invariance and differential item functioning (DIF), and score linking. I built its simulation: pooling platform scores at face value gives the wrong sign of daily aggregate sentiment on roughly a quarter of days, from measurement differences alone. For the application stage I fixed fifteen policy anchor events in advance and recorded them before computing any counts. The paper was submitted to arXiv in October 2026. I now apply the same lesson in industry: at Artemis AI, where I am co-founder and CTO, we are designing model evaluation around human-labelled data rather than agreement between models.

## 2. A discovery from a CSH research group

I chose Chen, Kmetty, Iñiguez and Omodei (2025). **In one sentence:** by linking a survey of Hungarian internet users to the YouTube histories they donated, the authors show that people who comment on political videos are more polarized and more selective than people who only watch, so studies built on visible traces such as comments overstate how divided political audiences are.

**The question it left me with:** is the gap between commenters and watchers stable enough, across platforms, topics and time, that a small donated panel could be used to correct polarization estimates built from public comments at scale?

**Why it interests me:** it has exactly the structure of the problem I work on. Public comments are a large, cheap and selected instrument; donated data is a small, expensive and closer criterion. If the relation between the two holds steady, a few hundred donors could calibrate millions of public comments, much as a small hand-labelled sample anchors a machine-labelled corpus in score linking. If it does not hold, each platform and period needs its own panel, and knowing where it breaks is itself a finding about where polarization measurement fails. It also matters for narratives: what spreads in comments may not be what spreads in viewing.

## 3. A problem I want to explore

**Statement.** When is a construct labelled by a machine (sentiment, stance or a narrative frame) comparable across platforms, annotator models and time, and how should the differences be corrected before we track how narratives spread between platforms?

**Why it matters.** Work on narratives at scale and on polarization increasingly pools platforms and uses LLMs as annotators. Any difference in how often a frame appears on Reddit and on X, or this year and last, mixes real change with a platform that elicits expression differently and an annotator that reads it differently, and that annotator changes version under our feet. Recent work approaches this from several sides. Gaynor, Miler, Goel, Hoyle and Resnik (2026) find that legislators' ideal points differ across votes, floor speeches and tweets, with Republicans more conservative on Twitter; reading that as a change in expression rather than an artefact requires the scale to mean the same thing in each venue. Fan et al. (2025) remove a document's medium from its embedding by linear concept erasure, Licht et al. (2025) study how LLMs can measure scalar constructs, and Hoyle et al. (2021) showed that an automated metric can drift away from the human judgment it stands for. I want to work on the step before a cross-platform comparison: a test of whether the

comparison is licensed, and a correction when it is not. The idea sits between two springboard projects, *Understanding narratives at scale* and *Polarization and minority representation on online platforms*, and close to Alexander Hoyle's work on measurement and validation with NLP in social science.

**Why it is tractable.** (1) My simulation is public, has a known ground truth and runs in seconds; extending it so that annotator models and time are facets alongside platforms would show how many hand labels, and which anchors, are needed to detect a difference large enough to reverse a conclusion. That paper needs no new data. (2) I can build the corpus: posts around a set of policy events fixed in advance on several platforms, collected through official APIs, frames coded by several LLMs, and a stratified hand-labelled sample per platform, larger than my 200 and with a second coder. A G-study would then split variance into platform, model, prompt and time, with DIF tests for each frame. (3) Concept erasure gives a direct test of whether removing the platform signal also removes real differences. (4) The corrected and uncorrected measures can then be put into a narrative-spread analysis with network methods, to report when conclusions change. The psychometric tools are mature; what is new is using platforms and models as the groups. I run LLM labelling chains in production and know where they fail. What I would need to learn is the network science of diffusion, which is one reason I am applying to CSH.

**For whom.** For researchers who pool platforms or switch annotator models mid-study and need to know whether their comparisons hold. For public bodies, NGOs and journalists who read social-listening dashboards: I build pipelines behind such dashboards, and their numbers are rarely checked against people. And for anyone asking whether a narrative really moved from one platform to another, or only looks as if it did because each platform is measured differently.

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